
OOP Homework 2

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Design explanation document

3 Abstract Classes:

1. Person.

- Everything in the hospital has only one type of person. This class is that type. It has virtual display() function and every kind of person can use it to describe themselves.

2. StaffMember:

- Any person who works in the hospital is a staff member. This class has getBillingRate() pure virtual function and every staff member overrides with their respective amount.
- There are three types of staff members.

1. General Practitioner

2. Surgeon

3. Nurse

All of them have their respective classes along with their own payment amount and specialization fields.

3. Ward:

- Every ward in the hospital can admit patients based on different criteria so it has a pure virtual function canAdmit() and admit().
- There are three types of staff members.

1. ICU

2. General Ward

3. Surgical Ward

All of them have their respective classes along with their own payment amount and admitting conditions.

Appointment and Scheduling:

The scheduling class makes bookings, cancellations, admits and discharges patients. All of the scheduling is done in the format of an appointment class which holds all of the key details about the appointment such as the staff and patient name along with date, day, and time.

Reporting and Queries:

I have created this class to make filtrations, sorting and managing the data in a particular order by using lambda functions.

Billing:

This class holds all of the operator overloads needed in order to calculate and compare different bills of different patients. Every patient has its object created whenever billing is called for the patient.

Hospital:

This class holds an object of everything. It holds a reporter from Reporting and queries, a scheduler from scheduling, wards, staff members and patients. All of the operations are performed through this class.

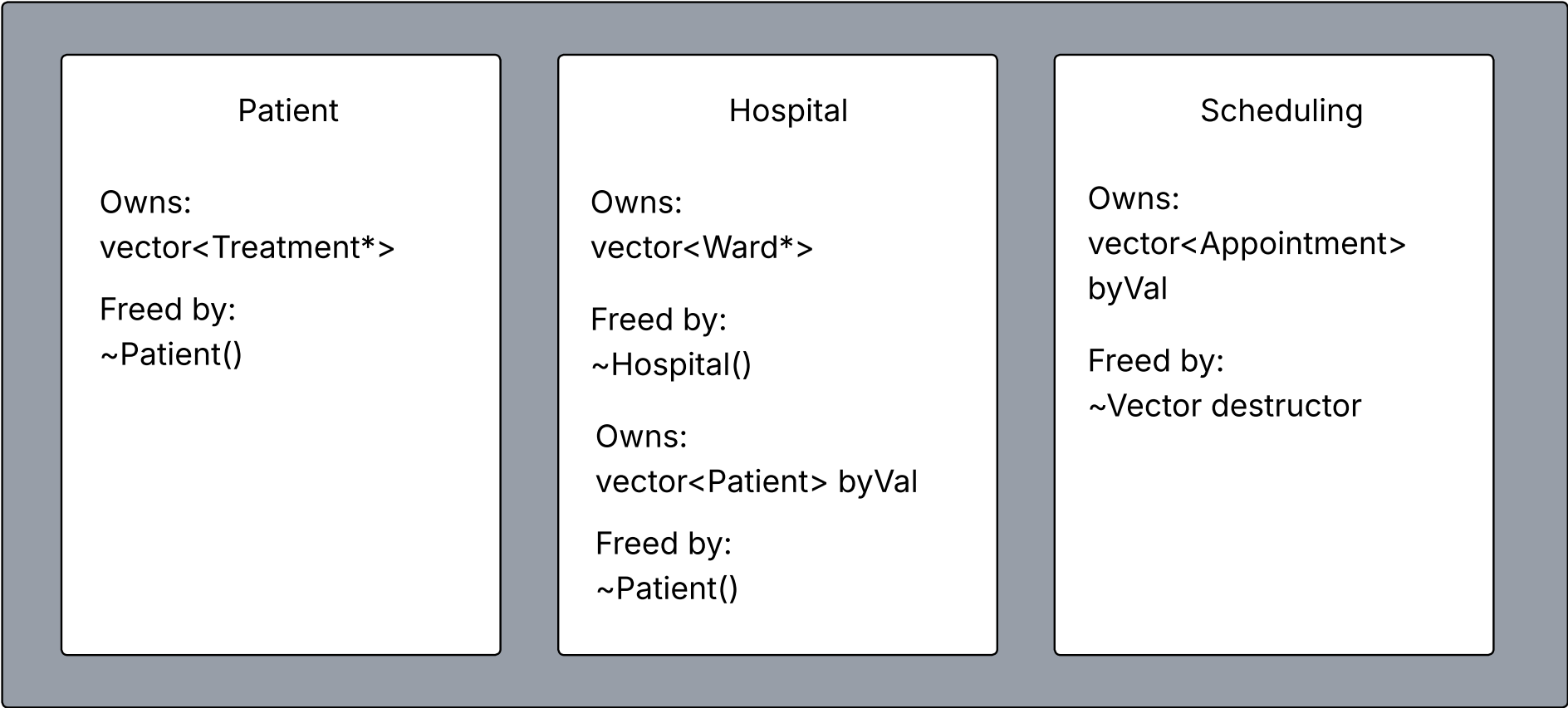
Memory Ownership Diagram

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Stack owned data



Heap allocated data



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